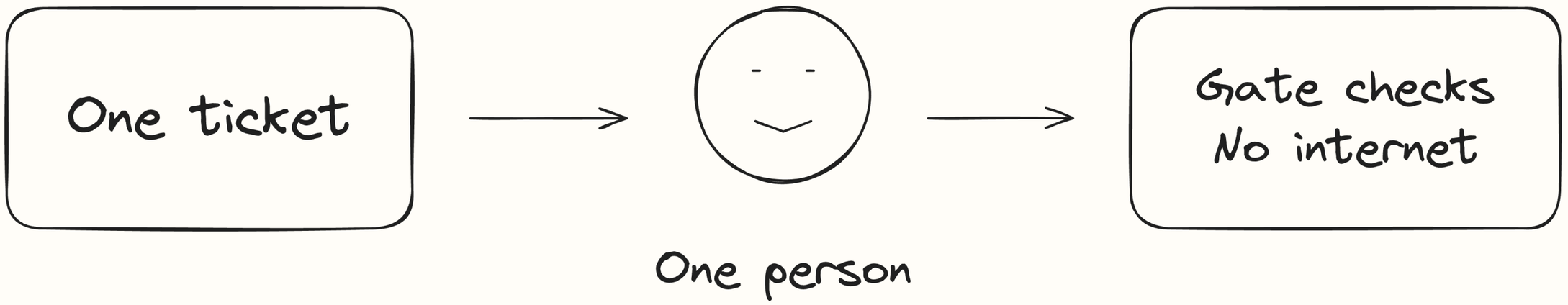
Private Pass



No central face database

A prepared prototype

QR ticket



A copy can be shared

Central face database



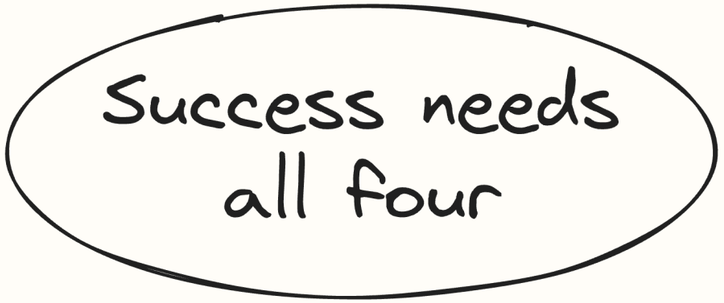
Lasting data at risk

Private Pass: link the ticket to its holder

Keep face checks on the phones

One person
One ticket

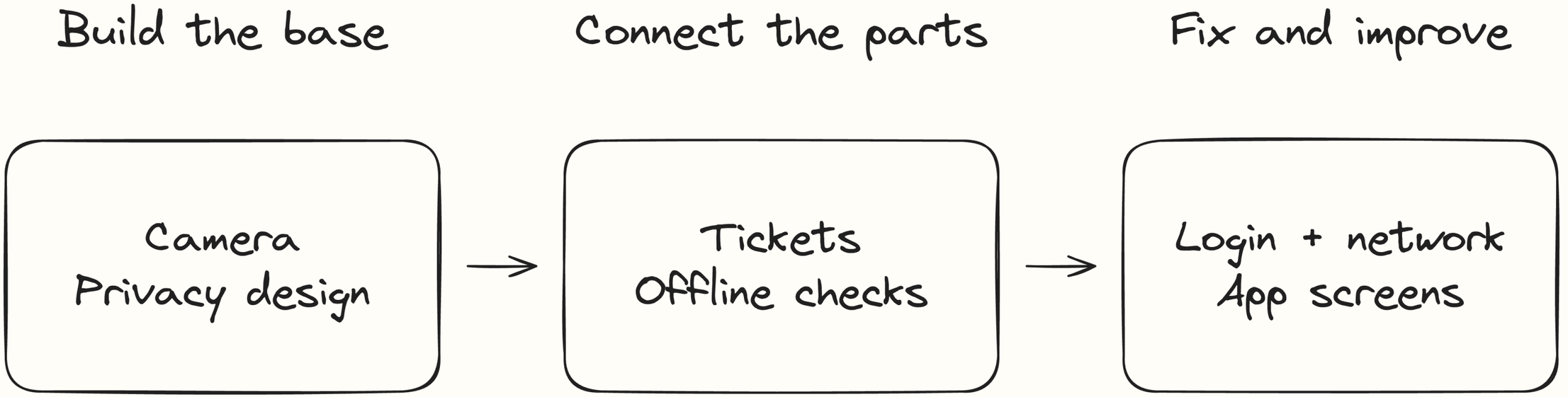
Prepared gate
No internet needed



No central reusable
face records

Saved tests
and honest limits

Supervisor: how does someone get a ticket?

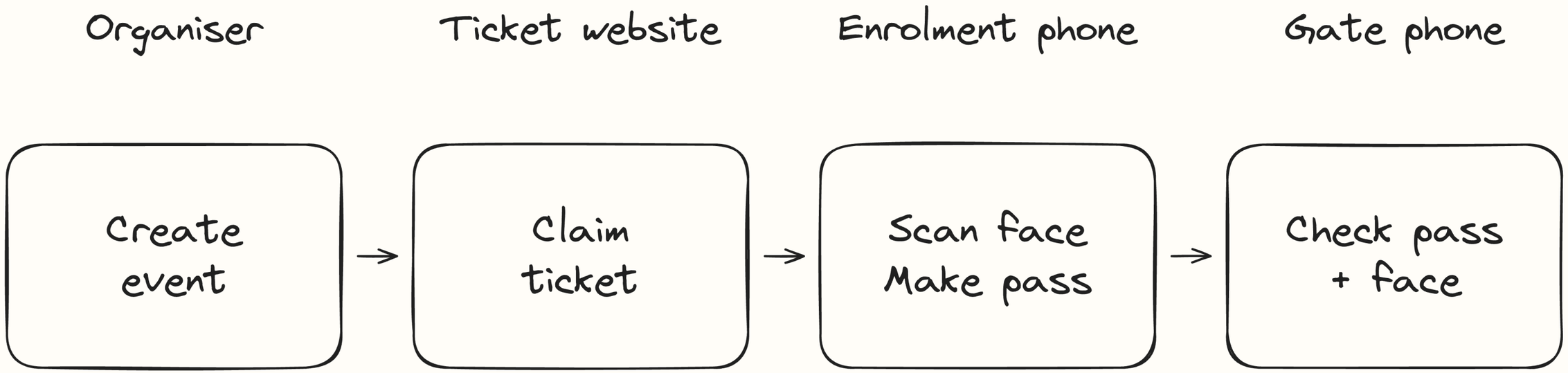


Adding ticketing made the journey complete

More features left less time for testing

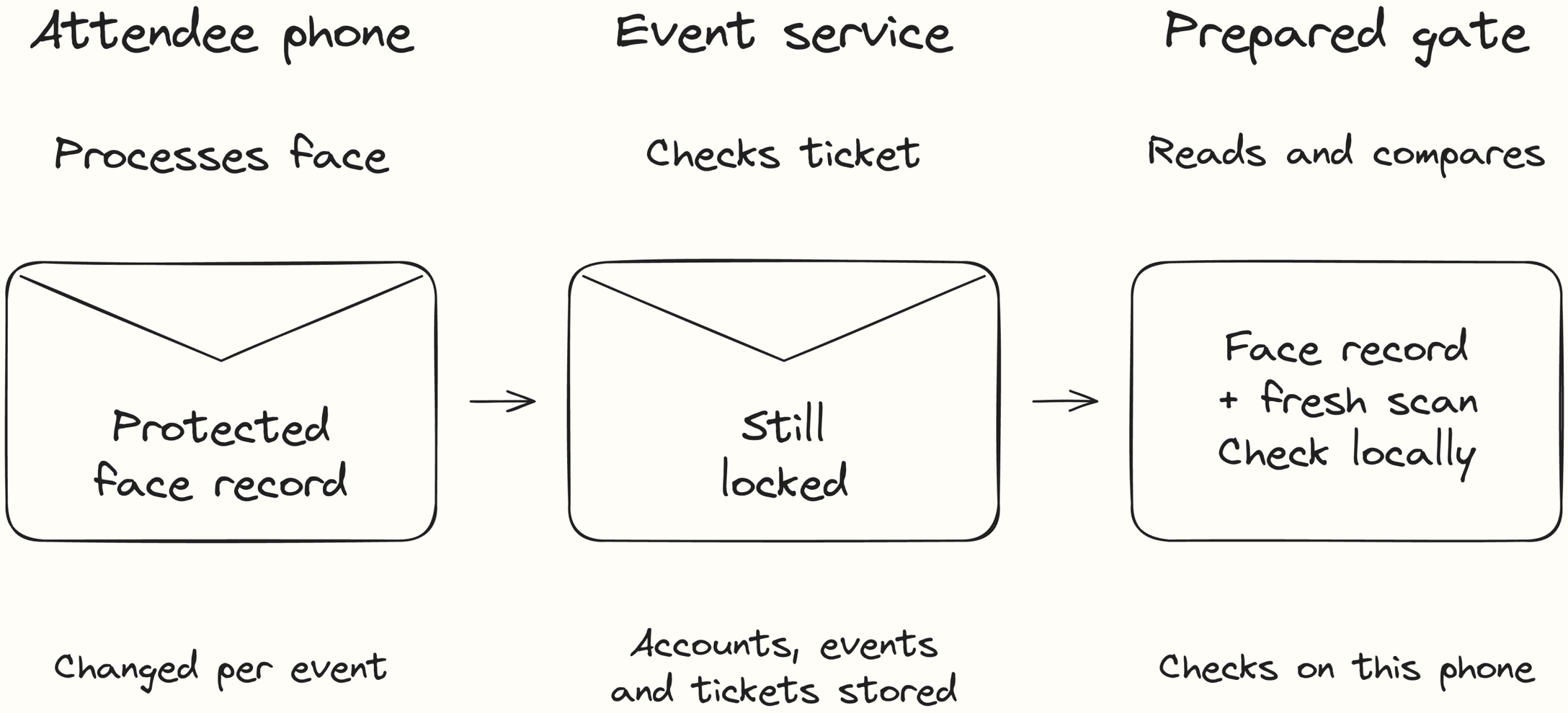
Faces can become comparable numbers	→	Compare one person with one ticket
Face numbers can still reveal information	→	Protect differently for each event
An issuer's stamp reveals changes	→	Check genuine passes without internet

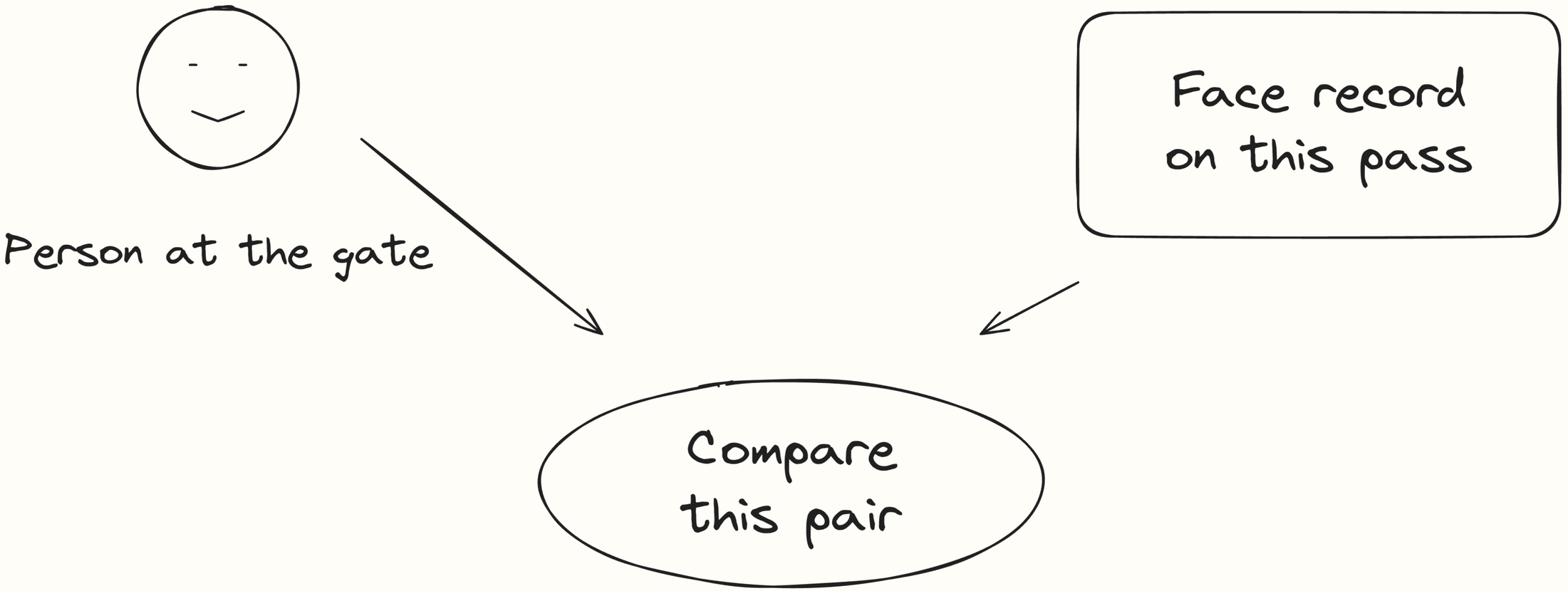
Research guides the design. It is not my test result.



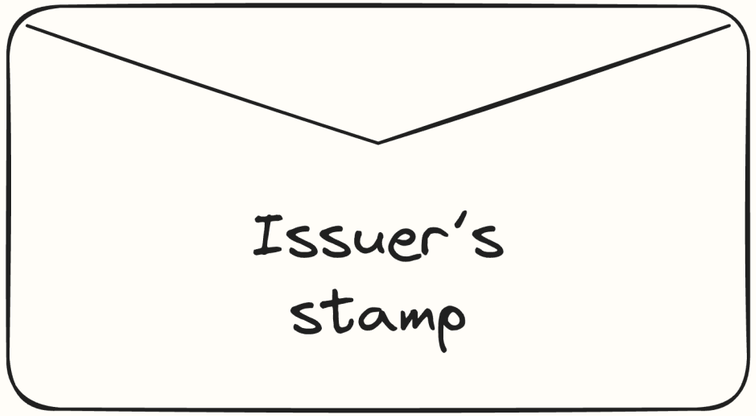
Earlier steps need a connection

Final check can work offline


Attendee phone
Processes face
Protected face record
Changed per event
Event service
Checks ticket
Still locked
Accounts, events and tickets stored
Prepared gate
Reads and compares
Face record + fresh scan
Check locally
Checks on this phone
Less central face data does not mean zero risk

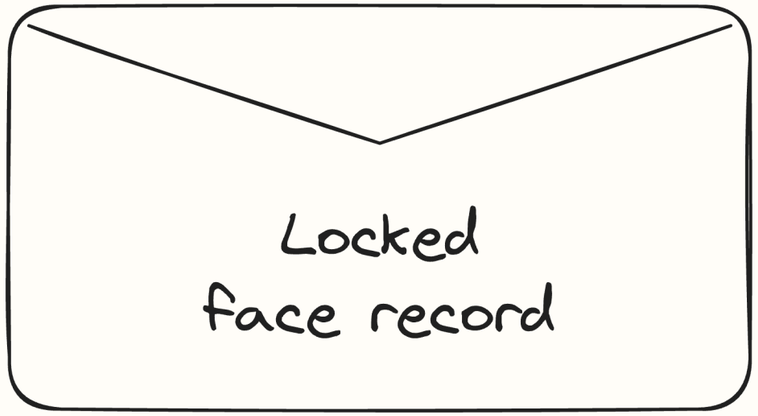

Person at the gate
Face record on this pass
Compare this pair
Too loose
Too strict
Wrong person gets in
Right person rejected

Signature



Genuine pass?
Has it changed?

Encryption



Who can read
inside?

A valid pass still needs a matching person

Prepare online, then disconnect

Right event + time
Genuine pass
Not cancelled or used
Matching face



Same pass again? Reject at this gate.

New cancellations wait for reconnection

Different face crops ⟶ Use the same crop rules

Reply lost. Tap again? ⟶ Same request, same ticket

App stops after success? ⟶ Save use before success

Saved checks
15 June

Ticket claims + access
Safe retries
Signatures + reuse
Sync + cancellation rules
iPhone launch + local Wi-Fi

Not established
on the final version

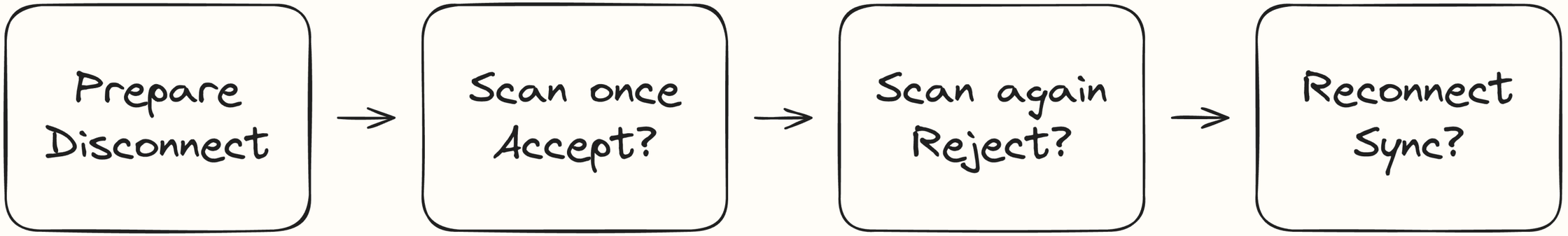
Detailed full-phone receipt
Face accuracy + fairness
Photo or screen resistance
Queue timings + user study

Bugs fixed: crop rules, safe retries, save before success

Software checks do not prove the full physical journey

Recorded	Software, browser, network checks Apps launched on an iPhone
Later observed	I saw the full journey work Detailed final record missing
Not measured	Face performance, queue speed User experience, public rollout

Planned demo: follow what actually happens



Download event information before disconnecting

A demonstration does not replace saved test data

Offline gate decisions	→	Partly evidenced
Stop ticket reuse	→	Supported by tests
Limit central face storage	→	Supported by design
Proven face performance	→	Not established

Cancelled while offline



Reconnect and refresh

Photo or screen shows a face



Test fake-face attacks

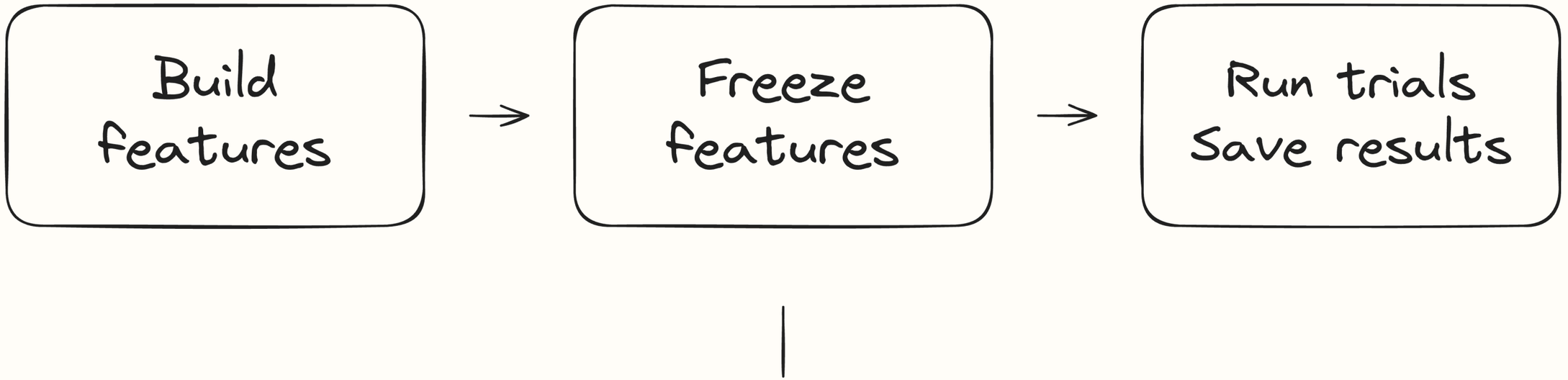
Cannot use a face check



Test another entry route

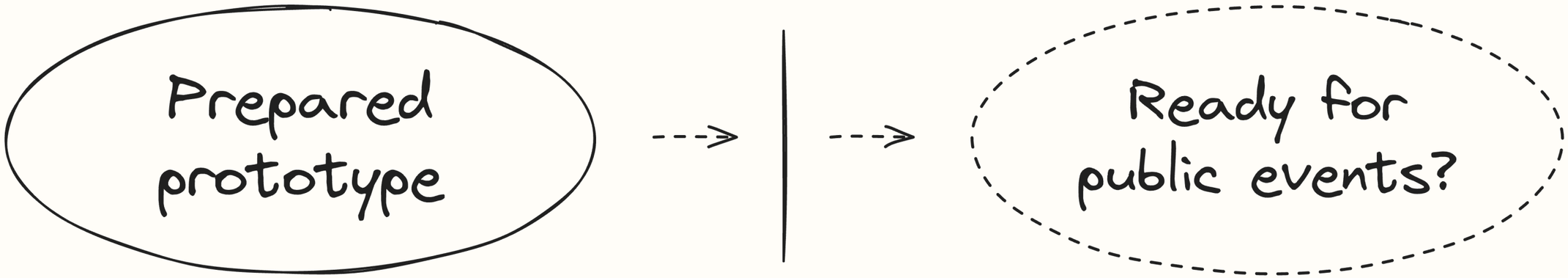
Needed before a real event

Next time



Set match rules first. Then test on new faces.

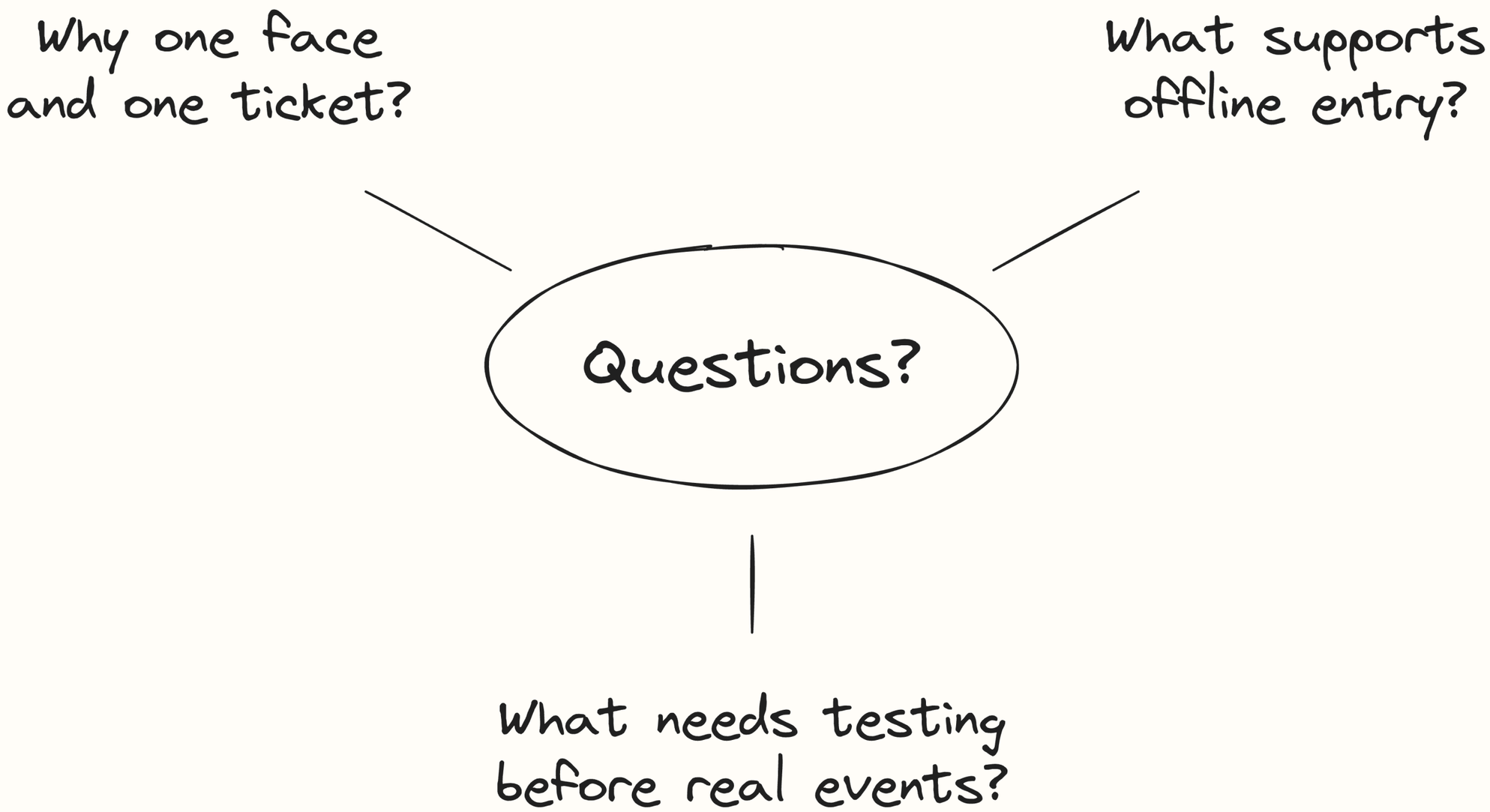
More proof needed



Ticket + face + offline gate
Can work together

Accuracy at scale?
Resistance to fake faces?

A working prototype is not a finished product


Why one face
and one ticket?
What supports
offline entry?
Questions?
What needs testing
before real events?